

15.1 Exam Review

Four modules, one stack. The exam asks you to connect a method to a measurement.

These notes match the lecture slides. Use **(slides)** on the course hub for the deck.

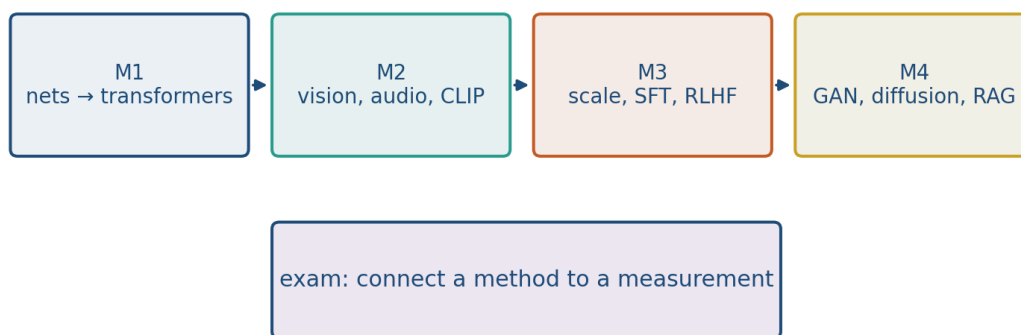
The exam is cumulative. It asks you to **name a method** and **say what you would measure**. Trivia without a metric is not the point of DATA 443/643.

1. Four modules

Module	Weeks	Spine
1	1–3	Neurons, embeddings, RNNs/gates, attention, GPT vs BERT
2	4–6	Multimodal, ViT patches, contrastive / CLIP / BLIP, audio as tokens
3	7–10	Scaling and efficiency, SFT and adapters, RLHF / DPO, safety and editing
4	11–14	GANs and mode coverage, diffusion and latent text conditioning, CoT / vote / faithfulness, RAG and ReAct

You should be able to draw the picture (two GAN players; $q(x_t | x_0)$; retrieve-then-generate) and attach **one number** to it (coverage, a denoising loss, recall@k, majority-vote accuracy, a faithfulness rate).

Four modules, one stack. The project sits on top of all four.



2. How to study

Rework the labs on CPU: XOR and embeddings, long-range gates, one attention head, toy CLIP retrieval, 2-D GAN coverage, one reverse diffusion step, vote-and-flag traces, bag-of-words RAG. Redo each note's three practice questions out loud. For every named method, write one sentence of the form "I would change X ; I would report Y on split Z ."

Formulas worth being able to write: a neuron, InfoNCE's diagonal story, attention softmax, GAN min-max, $q(x_t | x_0)$, one reverse mean, cosine for retrieval. You do not need to memorize paper years.

3. What the exam asks

Typical item: given a failure (collapsed modes, ignored chunk, lucky CoT answer, biased CLIP hit), **name the method that was in play** and **the measurement** that would have caught it. Another item: pick GPT versus BERT, CLIP versus BLIP, GAN versus diffusion, CoT versus ReAct, and justify with the task plus the metric.

Project language belongs here: baseline, one justified change, ablation, failure case. That is the same spine as note **14.3**.

Remaining talks

Canvas is official for who speaks when. If you still owe a presentation, bring a **failure case** (input, output, what broke), not only a demo that worked. Upload what Canvas asks. Time limits on the hub still apply.

4. Practice

1. A GAN's fakes look sharp and every one is the same face. What do you measure besides looking?
2. A RAG answer cites [d5] but the claim only appears in [d1]. Method name, and the check.
3. CoT accuracy is high; a calculator disagrees with half the traces. What exam idea is that, and which week's lab already scored it?